(3)(i) $\sin\theta = -\frac{\sqrt{3}}{2}$; $\operatorname{cosec}\theta = -\frac{2}{\sqrt{3}}$; $\sec\theta = -2$; $\tan\theta = \sqrt{3}$; $\cot\theta = \frac{1}{\sqrt{3}}$

(ii) $\sin\theta = \frac{\sqrt{5}}{3}$; $\operatorname{cosec}\theta = \frac{3}{\sqrt{5}}$; $\sec\theta = \frac{3}{2}$; $\tan\theta = \frac{\sqrt{5}}{2}$; $\cot\theta = \frac{2}{\sqrt{5}}$

(iii) $\cos\theta = \frac{\sqrt{5}}{3}$; $\operatorname{cosec}\theta = -\frac{3}{2}$; $\sec\theta = \frac{3}{\sqrt{5}}$; $\tan\theta = -\frac{2}{\sqrt{5}}$; $\cot\theta = -\frac{\sqrt{5}}{2}$

(iv) $\sec\theta = -\sqrt{5}$; $\cos\theta = -\frac{1}{\sqrt{5}}$; $\cot\theta = -\frac{1}{2}$; $\sin\theta = \frac{2}{\sqrt{5}}$; $\operatorname{cosec}\theta = \frac{\sqrt{5}}{2}$;

(v) $\cos\theta = \frac{5}{13}$; $\sin\theta = -\frac{12}{13}$; $\operatorname{cosec}\theta = -\frac{13}{12}$; $\tan\theta = -\frac{12}{5}$; $\cot\theta = -\frac{5}{12}$;

(5) $\theta = 60°, 120°, 240°, 300°,$

Exercise 3.4

(1)(i) $\sin(x+y) = \frac{220}{221}$ (ii) $\cos(x-y) = \frac{171}{221}$ (iii) $\tan(x+y) = \frac{220}{21}$

(2)(i) $\sin(A+B) = \frac{187}{205}$ (ii) $\cos(A-B) = \frac{156}{205}$ (3) $\cos(x-y) = \frac{4}{5}$

(4) $\sin(x-y) = -\frac{87}{425}$ (5) $\cos 105° = \frac{1-\sqrt{3}}{2\sqrt{2}}$; $\sin 105° = \frac{\sqrt{3}+1}{2\sqrt{2}}$;

$\tan\left(\frac{7\pi}{12}\right) = -(2+\sqrt{3})$

(7) $4x^2 - 2\sqrt{6}x + 1 = 0$ (16) 0 (22) 1 (24) $\frac{2}{11}$

Exercise 3.5

(1)(i) $\frac{161}{289}$ (ii) $-\frac{7}{25}$ (iii) $\frac{3713}{4225}$ (2)(i) $\frac{2\sqrt{3}}{5}, \frac{1}{3\sqrt{2}}$

Exercise 3.6

(1)(i) $\frac{1}{2}[\sin 63° + \sin 7°]$ (ii) $\frac{1}{2}[\sin 6x + \sin 2x]$ (iii) $\frac{1}{2}[\sin 12\theta + \sin 8\theta]$

(iv) $\frac{1}{2}[\cos 7\theta + \cos 3\theta]$ (v) $\frac{1}{2}[\cos\theta - \cos 9\theta]$ (2)(i) $2\cos 55°\sin 20°$

(ii) $2\cos 40°\cos 25°$ (iii) $2\sin 45°\cos 5°$ (iv) $2\sin 55°\sin 20°$

Exercise 3.8

(1)(i) $\theta = -\frac{\pi}{4}$; $\theta = n\pi + (-1)^n\left(-\frac{\pi}{4}\right)$. $n \in \mathbb{Z}$

(ii) $\theta = \frac{\pi}{6}$; $\theta = n\pi + \frac{\pi}{6}$, $n \in \mathbb{Z}$

(iii) $\theta = -\frac{\pi}{6}$; $\theta = n\pi + \left(\frac{-\pi}{6}\right)$, $n \in \mathbb{Z}$

(2)(i) $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ (ii) $x = \frac{2\pi}{3}, \frac{4\pi}{3}, \pi$

(iii) $x = \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$ (iv) $x = 0, \pi$